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Acknowledgements.

Stick some more acknowledgments here.

Abstract

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Intellectual Property

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Chapter 1

Introduction

1.1 Importance and Applications of Soft Robotics

Soft robots are a type of robot known for their use of soft materials, compliant nature and near infinite degrees of freedom . Degrees of freedom (DOF) is a term related to the number of independent movements a robot can make in 3D space.[2] Soft robots are usually nature inspired such as "vine-like" tip growing robots,[3,4] "snake like" continuum robots[5,6] and "hand-like" soft grippers.[7,8] The advantages of using a body that is highly deformable and with near-infinite DOF means these devices can navigate tortuous environments with ease.

This report focuses on continuum robots, often characterised by their long flexible bodies, commonly used for navigation and delivery or retrieval of objects in constrained environments.[9] Continuum robots can be classified into "continuous backbone" and "segmented back-bone" designs.[9] Continuous backbone devices are made from a soft and compliant material, in this report we will call these "soft-bodied", whereas, a segmented backbone refers to those that are made from hard, inflexible segments that are flexibly attached to achieve high curvatures, referred to here as "hard-bodied".

Hard-bodied continuum robots benefit from stiffer materials allowing a greater ease of

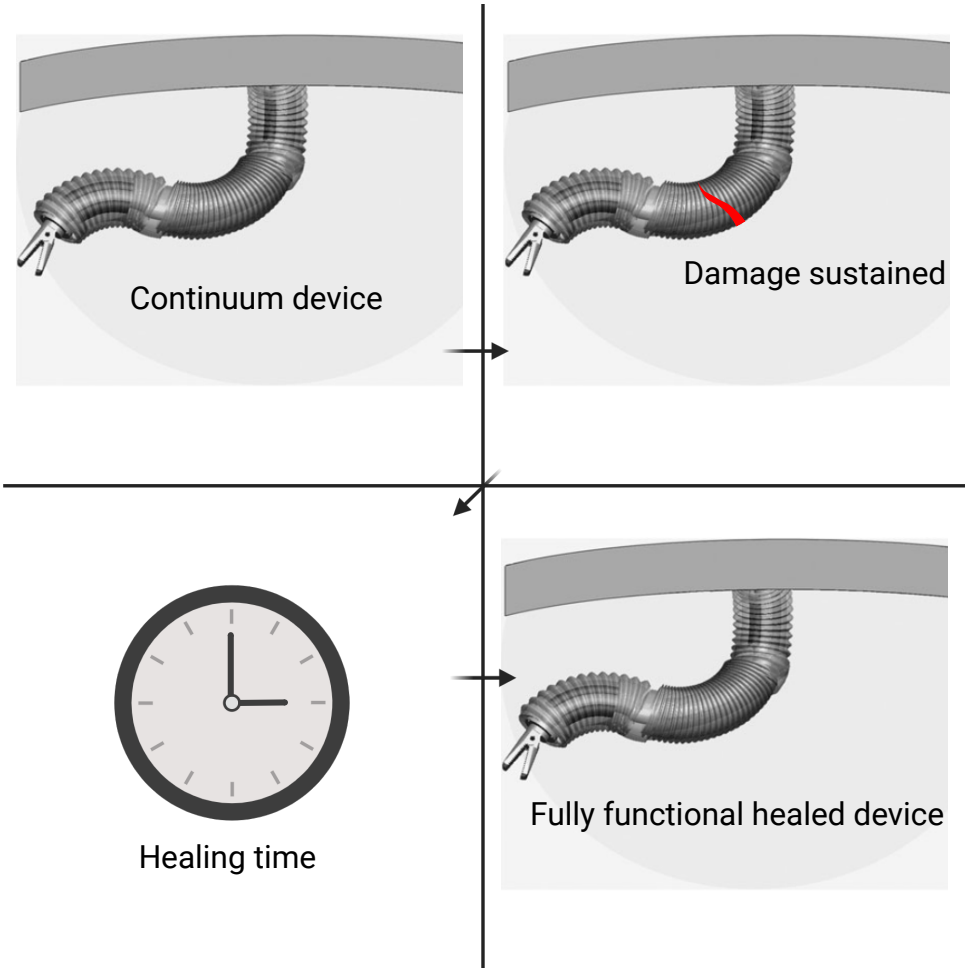


Figure 1.1: A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[1]

control[9] which can improve dexterity. However, they cannot deform and are more likely to impart a greater force on the surrounding environment.[10] Hard continuum robots are often tendon driven, whereby the segments are joined by wires that can be pulled to give specific movements. Whereas soft bodied continuum robots can achieve high curvatures and low forces on surroundings,[11] but are often harder to control due to their methods of actuation[9] and can suffer from material failure.[12] Soft continuum robots commonly take the form of an inflated body, these can be controlled in many ways such as varying pressure,[3] strain limiting a specific region[13,14] or magnetic actuation.[4,10] Pressurised systems that rely on fluids need a constant seal to maintain function as opposed to a segmented tendon driven approach that can be partially damaged and still actuatable. This leads the development of resilient materials such as self-healing polymers that can be applied to soft robotics.

1.2 Types of Self-Healing Polymers

Polymers are a type of molecule that consists of several repeating sub-units, these form chains that are combined many times over to form materials. Self-healing (S-H) polymers are a type of polymer that can heal damage.[15] This occurs through reactions and interactions between multiple polymer chains to reconnect chains that have been shifted apart through damage. The process of damaging or cutting a polymer does not break bonds in the polymer backbone but instead can break interchain interactions called cross-linking, allowing the chains to drift apart, reforming the cross-linking interactions will "heal" the material.[16]

S-H polymers can be classified into intrinsic or extrinsic healing, and autonomous or non-autonomous.[17] Materials with an intrinsic healing capacity involve a healing mechanism through chemical bonds or interactions that are continuously present in the polymer, rather than extrinsic healing, such as capsules of "healing" material that are broken through damage and causing the material to reform.[18] Autonomy refers to the healing

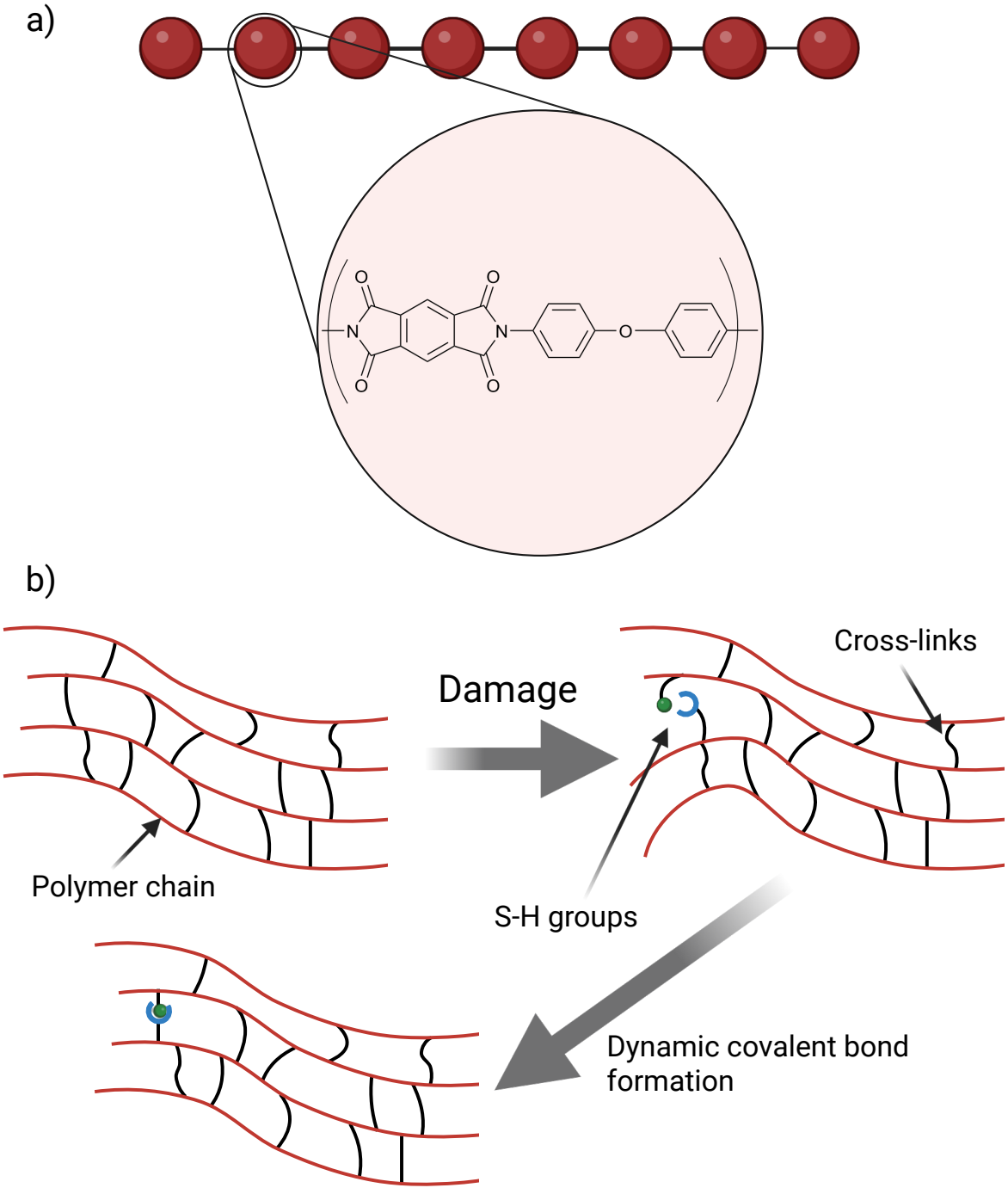


Figure 1.2: a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer

process occurring without external input. For example, capsules that burst upon damage of the material and release a healing agent are classified as autonomous, whereas a photochemical reaction that needs UV light to activate[19,20] is a non-autonomous mechanism.

1.3 Comparison of Different Self-Healing Pathways

There are a variety of different pathways that can be taken to create a S-H polymer depending on the chemistry of the initial polymer. The pathways that will be focused on are intrinsically self-healed types, however, both autonomous and non-autonomous types will be considered.

The three pathways below rely on a property of some electrons called conjugation. Conjugation occurs when specific electron orbitals overlap in such as way that allows the electrons to move around the molecule. This is often exploited with a property of some atoms called electronegativity, having a highly electronegative atom in a molecule will attract conjugated electrons to create a local high density region of electrons and therefore a polarisation in a molecule.

Dynamic covalent bonds can form and break based on the conditions applied to them[21] and they are ideal for use as mechanisms for S-H due to their intrinsic nature and reliability in production of high yields.[22] This report investigates the use of two types of dynamic covalent bonding, Diels-Alder and Michael additions, and an example of a supramolecular interaction. Supramolecular chemistry is based on interactions that are non-covalent and may be more suited to a soft material.[23] This is due to the S-H mechanism of the cross-links that forms interactions therefore the polymer chains can move semi-independently.[23,24] Whereas, a dynamic covalent bond has a much higher bond energy[21] and can therefore get closer to the original physical properties of a polymer.

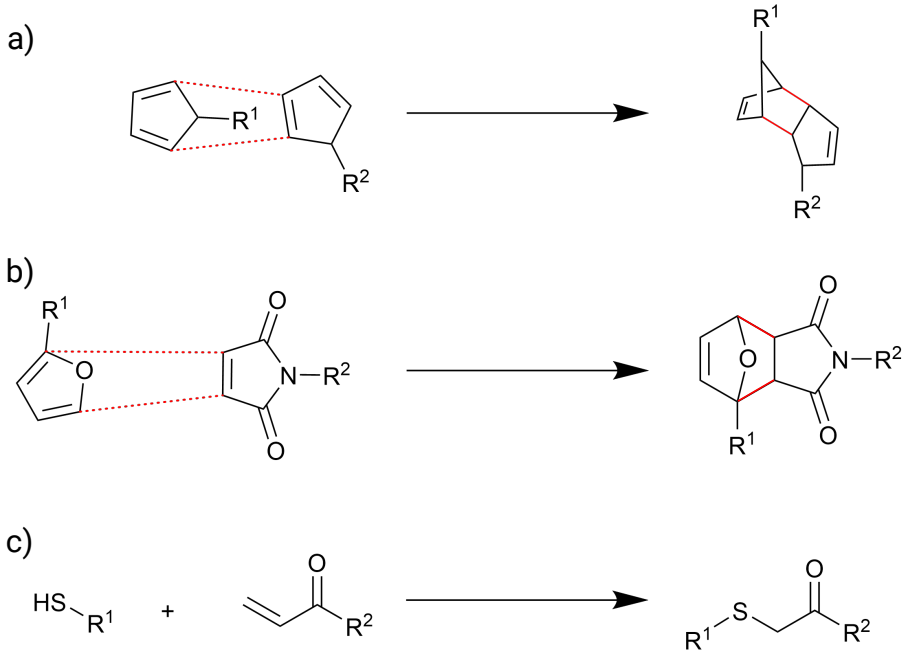


Figure 1.3: a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene.

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway

Diels-Alder (DA) reactions require a diene, a class of molecule characterised by having two sets of double bonds, and an alkene ("dieneophile").[21,25] The reaction occurs when conjugated electrons from the diene are free to move to form new bonds with the carbons either side of the alkene double bond, this creates a new ring structure. A faster reaction is generally observed with an electron rich diene and an electron poor dienophile.[26] The electronic character of the reagents can be altered by changing their local substituents. This class of reactions can have lots of different conditions depending on the molecules involved. Some variations can occur at room temperature,[27] which may make them an ideal choice for an autonomously healed medical device as the healing can occur whilst deployed in the body. However, many DA reactions are light[19,20] or heat activated.[28]

Some molecules such as cyclopentadiene can perform a DA reaction with itself, this has the potential to be an ideal choice for a S-H polymer as the molecule will never be in close proximity to a molecule that it cannot react with. However, some evidence has shown

that the bonds formed are not very strong and therefore high tensile strength materials may not be possible.[25]

One of the most common types of DA reaction is a heat activated reaction between a derivative of both furan and maleimide.[29] They react extremely accurately and require mild conditions,[25] therefore they are very suitable for application as a robotics material. One of the reasons that this reaction is so effective is that the maleimide has electron withdrawing carbonyl groups which make the dieneophile electron poor and therefore reacting faster. However, this mechanism has a disadvantage compared to cyclopentadiene as there are two halves of the end product. Therefore, there must be a two different processes occurring to create both the furan substituted and maleimide substituted chains. This increases the number of total steps in the reaction which increases the risk of low reaction yields and wasted material.

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway

Thiol Michael additions are a type of reaction that relies on a thiol, a molecule characterised by a sulfur containing group and a conjugated alkene.[30] The conjugated alkene used in a Michael addition is with a carbonyl group that will attract the electrons from the alkene to allow the thiol group to react. This has a similar advantage to the Diels-Alder reaction that it is not totally dependant on specific reactants merely groups of reactants. Therefore slight changes can be made to the adjoining groups to tweak the reaction conditions. This has been previously shown to have been successful with a heat activated reaction that could heal scratches after 15 to 30 minutes at 60 °C.[31]

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway

Pi-Stacking is a type of intermolecular interaction whereby conjugated systems interact based on the neighbouring substituents determining if a region is electron rich or poor.

These electron disparities cause an electrostatic interaction. This method commonly uses pyrenyl groups, these are polycyclic aromatic compounds, a very pi-electron rich species. These can be interacted with a napthalene diimide complex, an electron poor pi-system to give four pi-stacking interactions. A method has shown the creation of a polymer chain end group which can sandwich a napthalene group within two pyrenyl groups.[32] This method can be combined with other intermolecular interactions like hydrogen bonding[33] or mettalosupramolecular interactions to give a stronger interaction.[34]

1.4 Applications of Self-Healing Polymers in Soft Robotics

Continuing the theme of biological inspiration of soft robotics, S-H polymers can be used to extend the lifespan of components. In soft continuum robots, the goal is usually to make the robot softer than the environment surrounding it.[11] This presents a challenge that the environment can often be abrasive and can damage the device. It is therefore evident that a damage resilient material would be useful. Damage resilient materials can take multiple forms: Damage isolation, material elasticity, physical self healing such as low melting temperature materials and chemical self healing.[35]

1.4.1 State of the Art Self-Healing Soft Robots

State-of-the-art approaches using intrinsic S-H mechanisms are commonly based on DA chemistry.[17] These have been successfully implemented in tendon-driven[36] and pneumatically controlled soft robotic devices.[37,38] Such devices are capable of recovering from macroscopic damage, including punctures and cuts. Additionally, there are non-covalent approaches that utilize hydrogen bonding in the development of soft self-healing actuators, such as in McKibben muscle designs.[39]

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics

Terryn *et al.*[17] states that there are 5 main criteria for a S-H polymer; macroscopic damage must be healed completely; healing can take place multiple times at the same location; there are minimal material property changes after healing; elastomeric behaviour is achieved; the reprocessing and recycling of a material is possible. Here we propose that elastomeric behaviours are non-essential and a more useful requirement would involve a timely S-H reaction.

1.5 Limitations with Current Materials for Soft Robotics

Damage control whilst a device is active can be a challenge,[40] this shows a need for autonomous S-H.[17] Non-autonomous S-H can be made "functionally autonomous" through the use of stimuli responsive mechanisms, however, the current options are slow in operation.[40] Currently recovery timescales of applied S-H polymers are around 25 hours.[35,38] By altering the polymer used or S-H pathway this could be improved. There is also a lack of an easy synthesis route for many of the current polymer options,[35] thereby increasing the cost and reducing widespread use. Also the number of damage-repair cycles in these devices are low, such as 3 or less.[38] Most devices do not achieve all 5 criteria laid out in the Terryn *et al.* review,[17] such as the McKibben design of A. Lopez Diaz *et al.*[39] only reports a healing efficiency of 65%.

Chapter 2

Project Aims and Objectives

Chapter 3

Synthesis of a Self-Healing Polymer

3.1 Results

3.1.1 Hydrolysis of the Polyimide

The hydrolysis of the polyimide follows a standard mechanism for hydrolysis of an amide to a carboxylic acid using sodium hydroxide (NaOH). These show the hydroxy group attacking the carbonyl which then breaks the C-N bond and gives the carboxylic acid free.

Structural Analysis

Structural determination was conducted via Infrared spectroscopy. A comparison was made between the attempted hydrolysis of the sheets and a purchased form (polyamic acid).

The infrared spectrum of the hydrolysis attempts indicate that the crosslinking was not a successful reaction as the peaks in the spectrum are extremely similar to the unreacted form of the polymer. Also, in a comparison to the purchased PAA, which should be the

hydrolysed product that we were hoping for, the region around 3000 cm^{-1} is much busier and has the characteristic wide OH region. This would indicate that the hydrolysis hasn't occurred. This maybe due to the imide ring breaking and reforming as the temperature of the reaction is likely higher than the imide formation temperature meaning that the amide bond may get broken but in fact reforms at the high temperature. This seems a little unlikely though as I think there would be a suspension of material along with the main polymer.

The amount of NaOH was chosen to be in the ratio of 1:4 NaOH: monomer as there was a concern that we may fully hydrolyse the polymer (at all 4 amide bonds in the two imide rings) and give a tetrasubsituted benzene ring.

3.1.2 Spectrophotometry

The test for carboxylic acids as laid out by GS Johar would in theroy be a good test for determination of the hydrolysis of the polyamic acid. Therefore we tested for its interaction with acetic acid as the assumed positive test, along with DMSO as this was the solvent the polyamic acid was dissolved in, the polyamic acid in dmso, water and hydrochloric acid to test if ther reaction was purely just pH based. This was tested with three different concentrations of dyes. Only the results from the least concentrated samples will be included as the absorbance is too high for most of the analysis on the more highly concentrated versions.

In the literature, upon mixing of the eosin Y and Methylene blue "[there is an immediate colour change to green]". however our results determined that the colour was initially blue.

The absorbance of the sample with just water should give a baseline reading of how the absorbance of the dyes are. For a conclusive test, we should see the interaction between the dye mixture and the polyamic acid, should be similar. We see the flat region in the control test, between 600 and 700 nm grows in a similar shape when both acetic acid

and PAA are added. However, there is also a similar shape from the sample containing DMSO. This indicates that the test may not be very effective. A repeat of this test could be performed with a less concentrated dye mixture and the PAA in a different solvent as most of the peaks in the PAA scan are in the DMSO. A wider range could also be used as there is high absorbance in the low region (below 300nmn) that is not visible.

note: although the paper states to use Eosin Y, there is a common misnaming and the Eosin Y disodium salt must be used.

From the UV-vis data, it can be determined that the GS Johar method is not appoopriate for a determination of hydrolysis.

3.1.3 Cross-linking of the Polyimide

Cross-linking attempts follow a mechanism as proposed by (put reference here), where ethylene carbonate is used to form an ester with the carboxylic acids on neighbouring polymer chains, thereby joining them together with a "cross-link". This was performed with three different salts ($Et_4NBr, ZnCl_2, FeCl_3$) as catalysts.

During the cross-linking reaction with Et_4NBr, an orange-brown suspension was formed that when washed, filtered and freeze dried gave a pale orange powder. With $ZnCl_2$, the liquid was a dark orange clear liquid that, when water was added, gave a single clump of grey polymeric residue. This was freeze dried to remove the excess water and then analysed. The cross-linking reaction with the $FeCl_3$, gave a dark orange/brown gel, which gave a grey gum-like texture in small pieces when water was added. This was also frozen and analysed.

During a cross-linking with ethylene carbonate, CO_2 is released. This was a concern as the monowave reactor has a pressure limit of 20 bar. To ensure there was a reliable safety buffer, we set an artificial limit of 5 bar. The calculation to ensure this limit was kept used PV = nRT. (insert calculations).

From the IR data, the crosslinking with Et_4NBr was unsuccessful as the product didnt have the key peaks of the C-O bond in the ester or any alcohol peak. The alcohol peaks would be likely as the crosslinking was unlikely to be 100% successful and there would likely be the single ester product formed. However, neither of these were present and since the product was insoluble, no NMR could be obtained. This IR specturm was compared with the spectrum of Kapton (i.e. the ring closed form of the polyamic acid) and the similarities confirmed that the imide ring reformation had occured. The absorbance of the synthsised form from the PAA has much stronger absorbances. This is likely due to the fact that the clamping force on the kapton film is not as strong as in the powder and the powdered material is thicker so will absorb more just due to the material.

From the IR data of the ZnCl2 and the FeCl3, there are two new peaks at around 1800 which are not in either the kapton or the polyamic acid. These are likely the C-O bonds in the ehtylene carbonate starting material. There is also a new peak at 4.43ppm in the NMR spectrum, however, when adding 2 μL of EC in a "spike test", the peak increased in size, confirming that the new peak was just starting material. This

3.1.4 Moulding of Polyamic Acid

The moulding without the polymer gave a yellow powder, this is likely due to the boiling point of the NMP being higher than the reported cure temperature of the polyamic acid. Therefore the polymer cured whilst still in solution and therefore didn't form a fully stable cure.

However, the attempt after a week of drying was much more successful and the top layer gave a glassy smooth layer whilst layers below were still not cured in one piece. This is likely due to the polyamic acid forming a skin on top of the mould. This is due to the way that the smartvap works, the smartvap pumps a vortex of air over the surface of the material. This means that the top layer dries out much faster than layers below. This is fine when the sample in solution is small, however polymeric materials behave differently.

3.1.5 Catalytic Imidisation of polyimide

During the attempted crosslinking reactions, we observed the imide ring formation at a lower temperature than in the literature. We attempted to recreate this with the polyamic acid and the $ZnCl_2$ after observing that the crosslinking attempts would form the kapton at lower temperatures than expected.

The reaction without the catalyst was as expected, the material got slightly darker, however we attribute this to some of the solvent evaporating and therefore the substrate becoming more concentrated. When comparing the spectrum to the polyamic acid on its own, we determined that [nothing had happened?]. However in comparison to the perparation with $ZnCl_2$, the IR spectrum showed a spectrum more similar to the kapton. The spectrum has extra peaks at (vals) which match the spectrum from the polyamic acid. This likely shows that the reaction was incomplete. Further purification and analysis is neccessary

3.2 Experimental

3.2.1 The Hydrolysis of a Polyimide

Poly..... (Kapton) film (0.2g, thickness of x) was cut into pieces of approx. $3mm^2$, these were mixed with a solution of NaOH () in a reaction tube. This tube was placed into a monowave reactor that was set to ramp for 20 minutes and hold for 20 minutes at 100 °C. A sample was taken and the remaining mixture was reacted again on the monowave for ramp 20 mins and hold 20 minutes at 100°C. There was no observable difference in the Kapton film. Then this was thoroughly washed with water and IR was collected.

—table contents—
Table 3.1: table 1

3.2.2 Spectrophotometry as a Determination Method for Hydrolysis

Samples for Spectrophotometry analysis were made as in table 3.1.

A mixture of Eosin Y disodium salt and methylene blue was created as in the GS Johar test. THis was also diluted to with 5 parts water and 10 parts water.

Using a spectrophotometer, the samples (2mL) were tested between 250 and 750nm.

3.2.3 Crosslinking of Polyamic acid

This procedure outlines the crosslinking procedure of Polyamic acid via ethylene carbonate using a catalytic salt as outlined in previously published procedures.

3.2.4 Cross-linking with TEAB and Ethylene carbonate

A mixture of polyamic acid (0.51mL, conc) was mixed with ethylene carbonate (0.08g) and tetraethylammonium bromide (). This was mixed in a monowave reactor set to ramp for 20 minutes and hold for 1 hour at 150°C. This gave an orange suspension that was dried to give a pale orange powder as the final product (grams, yield).

3.2.5 Cross-linking with other salts

The crosslinking with ZnCl2 and FeCl3 was performed at 100°C for 1 hour. With ZnCl2 a clear dark brown viscous liquid was obtained and with the FeCl3, a clear dark brown gelatinous liquid was obtained. A solid was extracted from each of these liquids and

spectroscopic analyses were performed.

3.2.6 Spectral Data

tetraethylammonium bromide Crosslinking

A orange powder (grams,moles,yield) was isolated, ν max/cm^{-1}(solid);

Iron chloride Crosslinking

A white gum (grams,moles,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid);

Zinc chloride Crosslinking

Brittle brown pieces (grams,moles,yield) were isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid);

3.2.7 Moulding of Polyamic Acid

Polyamic acid in NMP (stats) was poured into a Cone-shaped mould. This was assembled with a 3D printed part to connect to the SmartVap (Supplier) for 1 week at 100°C. This reduced the volume and made the liquid a darker brown. This was heated to 150°C for 1 hour to cause imidisation before being extracted from the mould with a scalpel. This experiment was also repeated without drying off the solvent first.

3.2.8 Catalytic Imidisation of Polyamic Acid

Polyamic acid (0.51mL), $ZnCl_2$ (0.02g) and a magnetic stirbar were added to a Monowave tube. This was placed in the monowave at 110°C for 1 hour (ramp for 20 mins, hold for 60 mins). A control experiment without the $ZnCl_2$ was also placed under the same conditions. The control expeiment gave a clear dark brown viscous liquid which when dried gave a gum-like pale orange solid. The test with the $ZnCl_2$ gave a similar dark brown liquid however this was slightly cloudy and when processed, gave a orange powder.

Control Spectrum

IR(solid): vals, vals cm^{-1}

Catalytic Imidisation Spectrum

IR(solid): vals, vals cm^{-1}

Chapter 4

Construction of a Bellows Based Actuator

4.1 Design

Actuators were constructed using a combination of a folded polyimide sleeve and a internal silicon tube with nitinol wires for actuation.

4.1.1 Construction of a Folded Protective Sleeve

The polyimide film was folded into a []-style design as based on the previously published work. For the proposed design, the film selected was Kapton® (Supplier, thickness).

Manual folding versus using a plotter

A commercially available Cricut® was used with a scribe tool and a cutter to achieve a rapid manufacturing of a folded sleeve. This was attempted with paper initally for cheap prototyping and then transferring to the polymer film.

The paper was easy however the polyimide did not take up the scores so easily.

4.1.2 Complete Assembly

The complete assembly was formed with the polyimide sleeve, the silicon continuum device and 3D printed parts. The construction method is as follows: a sheet of polyimide is marked with the desired folding pattern (as shown in figure), the sheet is folded by hand before being shaped into a tube and glued along an overlapping edge. The folds are encouraged into place and one end is sealed. This is the assembled with 3D printed parts to form the actuator.

The silicon actuator contains 7 holes where three lengths of nitinol wire were threaded through and fastened with two grub screws each in 3D printed handles. This leaves a separate tube through the centre of the silicon tube that could be used in future work for targeted delivery of liquid or components such as naneye cameras for observation.

4.2 Methodology

Our methodology is inspired by similar "bellows-style" approaches to construct continuum devices (references) with the key difference in the use of Polyimide sheets (Kapton) forming the main body of the device.

This work focuses on the use of Kapton (thickness =) due to its good robustness and high temperature capabilities.

4.2.1 Stiffness Characterisation

THe stiffness of the polyimide sleeve applies an additional force on the silicon actuation device. For instance, when the device is pressurised, the additional forces applied to the

device from the stiffness of the sleeve will affect the amount of air pressure needed to vary the actuators length. To characterise the stiffness profile, a linear stage was used with a load cell (brand) held in a SmallRig Magic Arm. This was attached to the end of the polyimide sleeve which was clamped in a 3D printed stand. The whole experimental setup was attached to a ThorLabs Breadboard. The linear stage was set to 60mm (the resting length of the actuator was 120mm, the linear stage is in reverse so to get to 60mm, the linear displacement is 90) and the load cell was zeroed. The load cell was moved to 150mm (actuator length is 20mm compressed) and then the linear stage was set to go to 0mm at 0.5mm s^{-1}. This contained a compression force and a stretching force. The speed of the actuator was translated into the displacement using:

$$v = \frac{d}{t}$$

. The load cell data was collected using DSCUSB Toolkit and processed in python.

4.2.2 Temperature Stability of the Polyimide Sleeve

Using the same experimental set-up as the stiffness characterisation. The kapton® was put in an oven at 200°C for 1 hour, left to cool and then compressed and stretched as before.

4.2.3 Workspace Characterisation

4.2.4 Moulding of polyamic acid

Polyamic acid undergoes an imidisation reaction at 150-200° to form Kapton. It shoudl, in theory be prossible to pour the polyamic acid into a mould and heat to form Kapton. This was attempted by pouring the PAA into a mould and heating to 150°C for 20 mins to cause imidisation. This was repeated after placing half the mould into the smartvap

with a 3D printed adaptor. The smartvap heats the sample and blows a stream of air into a vortex over a sample to remove the solvent. This was performed for a week before then heating the sample to 150°C for 1 hour.

4.3 Results

4.3.1 Stiffness Characterisation

4.3.2 Temperature Stability of the Bellows

4.3.3 Workspace Characterisation

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